Mahavir Education Trust's
Shah & Anchor Kutchhi Engineering College
Autonomous Institute Affiliated to University of Mumbai
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Machine Learning

Academic Year: 2026-2027

# PCOSense

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# CHAPTER 1: INTRODUCTION

## 1.1 Introduction

Polycystic Ovary Syndrome (PCOS) is one of the most common hormonal disorders affecting women. Due to varied symptoms such as irregular menstrual cycles, acne, weight gain, excessive hair growth, and hair loss, PCOS often remains undiagnosed for years. Delayed diagnosis can lead to serious health problems including infertility, diabetes, obesity, and hormonal imbalance.

PCOSence is an AI and Machine Learning-based web application developed to predict the risk of PCOS at an early stage. Users enter their health details, and the system analyzes the data using a trained Machine Learning model to estimate the likelihood of PCOS. Along with the prediction, the system provides personalized diet, exercise, and lifestyle recommendations.

## 1.2 Problem Statement

Many women are unaware that they have PCOS because its symptoms are often ignored or misunderstood. Diagnosis usually requires multiple medical tests and specialist consultation, which may not be easily accessible. This delay increases the risk of long-term health complications. Therefore, there is a need for a simple, quick, and accessible system that can perform preliminary PCOS risk assessment.

## 1.3 Objectives

The main objectives of the project are:

- Predict the risk of PCOS using Machine Learning.
- Provide a quick preliminary health assessment.
- Recommend personalized diet and exercise plans.
- Raise awareness about early PCOS symptoms.
- Reduce delays in diagnosis.
- Develop an accessible and user-friendly healthcare platform.

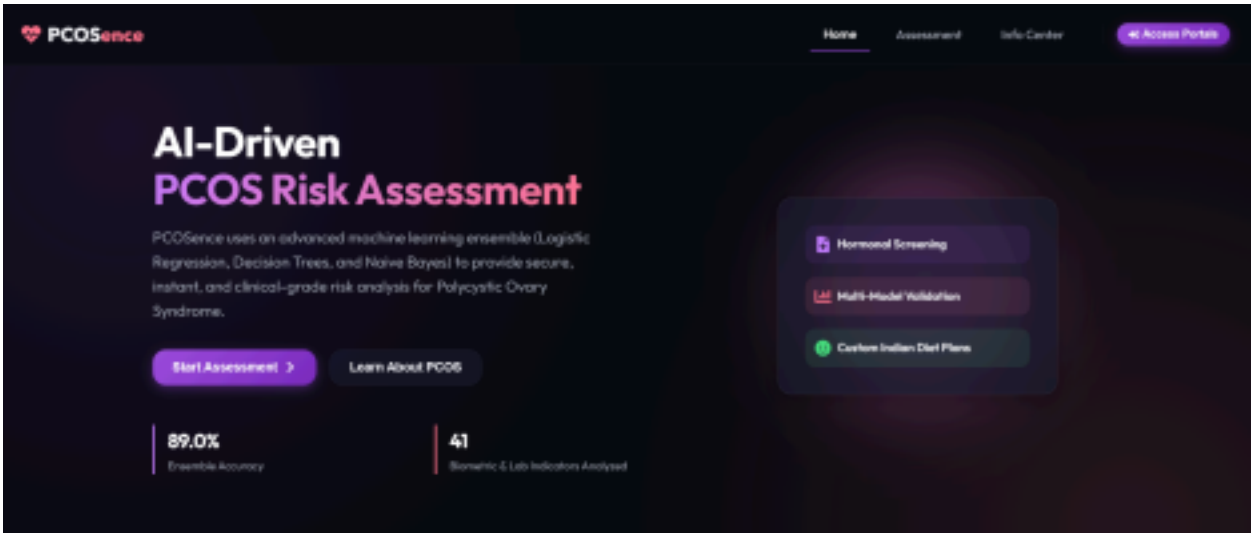
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# CHAPTER 2: SYSTEM OVERVIEW

## 2.1 Proposed System

PCOSence is a web-based application that allows users to enter their health information through an interactive interface. The entered data is sent to a Flask backend where the Machine Learning model processes it and predicts whether the user is at risk of PCOS. The system then displays the prediction result along with health recommendations.

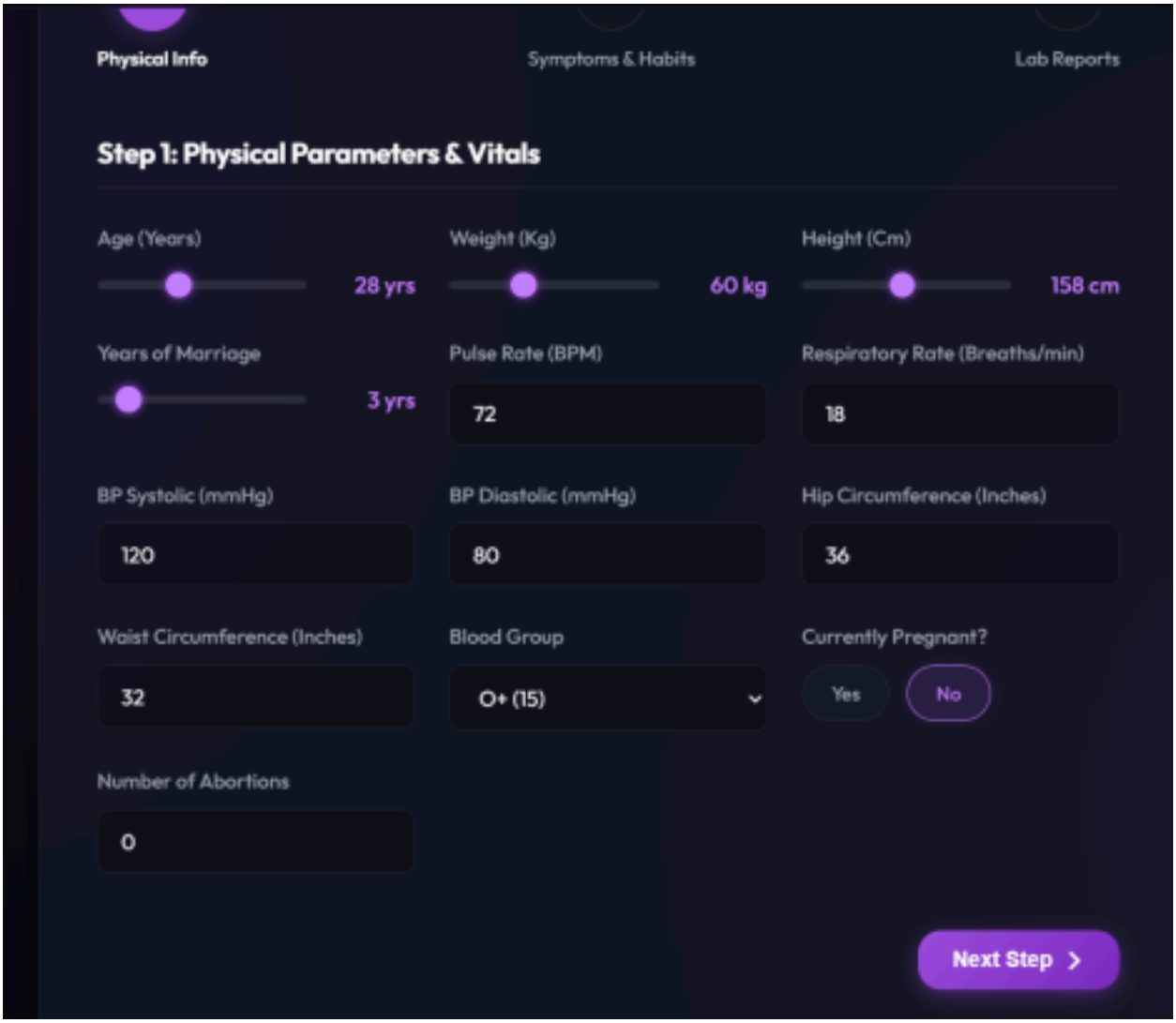
Figure 2.1 Home Page




Description:

The home page introduces users to the PCOSence platform and provides information about the purpose of the application. It contains navigation options and allows users to begin the PCOS risk assessment.

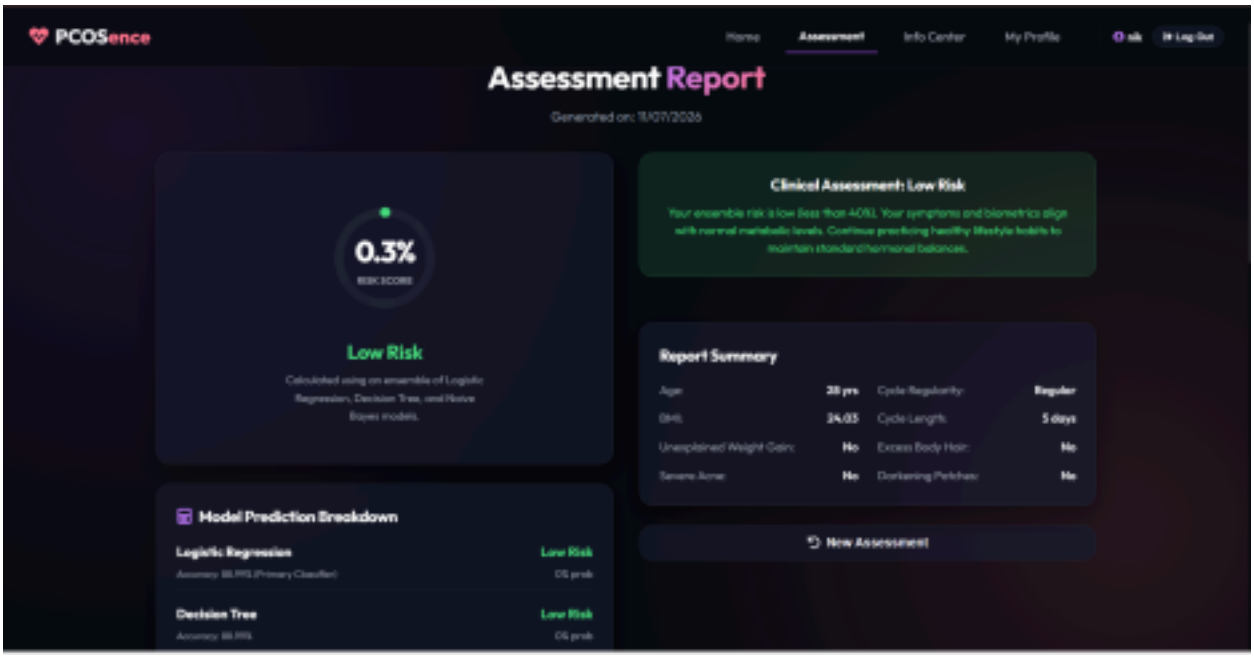
Figure 2.2 Prediction Form




Description:

The user enters health-related details such as age, BMI, menstrual cycle information, weight gain, acne, hair growth, exercise habits, and other lifestyle factors. These details are used as input for the Machine Learning model.

Figure 2.3 Prediction Result




Description:
After processing the input, the system displays the predicted PCOS risk along with personalized health recommendations, including diet, exercise, and medical consultation advice.

# CHAPTER 3: METHODOLOGY

## 3.1 Dataset

The project uses the Kaggle PCOS Dataset, which contains important health parameters such as:

- Age
- Height
- Weight
- BMI
- Menstrual cycle details
- Acne
- Hair growth



- Hair loss
- Weight gain
- Exercise habits
- Lifestyle factors

The target variable is PCOS (Yes/No).

## 3.2 Data Preprocessing

Before training the model, the dataset was preprocessed using the following steps:

- Removed duplicate records.
- Handled missing values.
- Selected important features.
- Normalized the data.
- Split the dataset into training (80%) and testing (20%) sets.

## 3.3 Machine Learning Model

Three Machine Learning algorithms were compared:

- Logistic Regression
- Decision Tree
- Naïve Bayes

After evaluation, Logistic Regression was selected because it provides a good balance between accuracy, speed, and interpretability.



# CHAPTER 4: SYSTEM ARCHITECTURE

## 4.1 Workflow

The working of the system follows these steps:

1. User enters health information.
2. Data is sent to the Flask backend.
3. Backend validates the input.
4. Machine Learning model predicts PCOS risk.
5. Results and health recommendations are displayed to the user.

# CHAPTER 5: TECHNOLOGY STACK

| Component | Technology Used |
|---|---|
| Frontend | React.js, CSS |
| Backend | Flask, Python |
| Machine Learning | Scikit-learn, Pandas, NumPy |
| Database (Future) | MongoDB |
| Deployment | Vercel, Render |

# CHAPTER 6: FEATURES

The main features of PCOSence include:

- AI-based PCOS risk prediction.
- User-friendly web interface.
- Personalized diet and exercise recommendations.
- BMI calculator.
- Downloadable health report.
- Mobile-friendly responsive design.
- Future hospital integration support.

# CHAPTER 7: ADVANTAGES

- Early detection of PCOS.
- Fast and affordable screening.
- Easy to use.
- Accessible from any device.
- Personalized health guidance.
- Encourages timely medical consultation.

# CHAPTER 8: FUTURE SCOPE

Future improvements include:

- Hospital integration.
- Online appointment booking.
- Nearby hospital locator.
- AI health chatbot.
- Period tracker.
- Menstrual cycle prediction.
- Medicine reminders.
- Wearable device integration.
- Cloud database.
- Multi-language support.

# CHAPTER 9: CONCLUSION

PCOSence is an AI-powered web application that helps women perform an early assessment of PCOS risk using Machine Learning. The system provides quick predictions along with personalized lifestyle recommendations, making early detection more accessible. Although it does not replace professional medical diagnosis, it serves as a valuable preliminary screening tool that encourages timely consultation with healthcare professionals.

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